ChatGPT

Mathematical Formulas and Decision Structures for the AI Decision-Support Skill (May 2026)

Bottom-line summary

Implementing an adaptive decision-support skill requires formalizing several decision-analysis methods into mathematical expressions. Key inputs include **candidate alternatives**, **criterion scores** and **preference weights**, while outputs are **ranked alternatives**, **eliminated options** and **explanatory metrics**. Deterministic formulas govern the scoring and elimination (e.g., weighted sums, distance metrics and outranking indices) [1] [2], while stochastic or adaptive formulas choose which questions to ask (information gain, expected regret) [3]. The pseudocode provided at the end outlines a generalized logic flow that can be tailored to diverse domains.

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1. Common inputs and outputs

1.1 Inputs

Symbol	Description	Example / Notes
$A = \{a_1, \ldots, a_m\}$	**Alternative set** – the options available to the user.	Houses, app feature sets, academic programs.
$C = \{c_1, \ldots, c_n\}$	**Criteria set** – attributes used to compare alternatives.	Price, distance, size, risk. Each criterion can be beneficial (higher is better) or cost (lower is better).
x_{ij}	Performance of alternative a_i on criterion c_j.	A numerical value or ordinal score. Missing values may be estimated or imputed.
w_j	**Weight of criterion c_j.**	Elicited via methods like PAPRIKA, AHP or BOED; $\sum_j w_j = 1$ for additive models.
$\mathcal{H}$	Hard constraints (veto criteria).	Budget limit, geographic boundaries. Options violating constraints are removed.
θ	User's latent preference vector / utility parameters.	Treated as random variables in Bayesian elicitation.
$p(\theta)$	Prior distribution over preferences.	Often Dirichlet or normal; updated as questions are answered.

Symbol	Description	Example / Notes
D	Candidate question set.	Each question asks for a comparison (which of two alternatives is preferred, or which attribute trade-off is more important).

1.2 Outputs

Output	Description
W	Final weight vector $(w_1, \ldots, w_n)$ representing relative importance of criteria.
$\mathbf{S} = (s_1, \ldots, s_m)$	Scores for each alternative computed by the scoring method (WSM, TOPSIS, etc.).
$\mathrm{Rank}(A)$	Ranking or ordering of alternatives based on scores.
Shortlisted set	A subset of A that remains after elimination (e.g., top 5 alternatives).
Justification report	Explanation of why top alternatives were selected, including weights and performance on criteria.
Updated posterior $p(\theta \mid \mathrm{answers})$	Distribution reflecting the remaining uncertainty after elicitation.

2. Deterministic scoring methods

2.1 Weighted Sum Method (WSM)

Purpose: Derive a composite score for each alternative by summing performance across criteria multiplied by weights [1].

Normalization: For benefit criteria, scores may be normalized as $\tilde{x}_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}$; for cost criteria, use $\tilde{x}_{ij} = \frac{\max_i x_{ij} - x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}$. This ensures all $\tilde{x}_{ij} \in [0, 1]$.

Score computation:

$$s_i = \sum_{j=1}^{n} w_j \cdot \tilde{x}_{ij}.$$

Ranking: Alternatives are ranked in descending order of s_i. The method is simple and transparent but assumes full compensability (poor performance on one criterion can be offset by excellent performance on another).

2.2 TOPSIS (Technique for Order Preference by Similarity to Ideal Solution)

TOPSIS ranks alternatives by their closeness to a **positive ideal solution** (best values on each criterion) and distance from a **negative ideal solution** (worst values) [2] .

Normalization: Use vector normalization: $r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{k=1}^{m} x_{kj}^2}}$. For cost criteria, invert the sign or use appropriate transformation.

Weighted normalized matrix: $v_{ij} = w_j \cdot r_{ij}$.

Ideal and anti-ideal points:

$$v_j^+ = \max_i v_{ij}, \quad v_j^- = \min_i v_{ij}.$$

Distances: For each alternative a_i, compute Euclidean distances to the positive and negative points:

$$d_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, \quad d_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2}.$$

Closeness coefficient:

$$C_i = \frac{d_i^-}{d_i^+ + d_i^-}.$$

Ranking: Rank alternatives descending by C_i. A higher C_i indicates closeness to the ideal and remoteness from the anti-ideal.

2.3 Outranking (ELECTRE I/III simplified)

ELECTRE compares alternatives pairwise using **concordance** and **discordance** indices. Let a and b be two alternatives.

Concordance set: $J_{ab} = \{j \mid v_{aj} \geq v_{bj}\}$ where v_{ij} is the weighted normalized score on criterion j.

Concordance index:

$$C(a,b) = \sum_{j \in J_{ab}} w_j.$$

Discordance index:

$$D(a,b) = \max_{j \in K_{ab}} \frac{v_{bj} - v_{aj}}{R_j},$$

where $K_{ab} = \{j \mid v_{aj} < v_{bj}\}$ and R_j is the range of criterion j (e.g., $\max_i v_{ij} - \min_i v_{ij}$). A discordance threshold d and concordance threshold c are set. Alternative a **outranks** b ($a \succ b$) if $C(a,b) \geq c$ and $D(a,b) \leq d$. ELECTRE III uses preference, indifference and veto thresholds to soften these conditions [4].

Net outranking flow: Summarize the net favourability of an alternative by subtracting the count/weight of alternatives it outranks minus those that outrank it. Alternatives with higher net flow survive to the next stage.

2.4 Analytic Hierarchy Process (AHP) weight derivation

In AHP, decision makers provide pairwise comparison ratios a_{ij} for the importance of criterion c_i relative to c_j. The reciprocal property requires $a_{ji} = 1/a_{ij}$. Let A be the comparison matrix.

Eigenvector method: Compute the principal eigenvector $\mathbf{w}$ solving $A\mathbf{w} = \lambda_{\max}\mathbf{w}$ and normalize so $\sum_j w_j = 1$ [5]. The consistency ratio (CR) measures reliability: $\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}}$, where $\mathrm{CI} = \frac{\lambda_{\max} - n}{n-1}$ and RI is a random index. CR < 0.1 is considered acceptable [6].

AHP-express and Tournament Tree Method: AHP-express reduces the number of comparisons to $n-1$ per criterion by using a linear estimation procedure [7]. The Tournament Tree Method (TTM) obtains a complete and consistent comparison matrix with only $m-1$ comparisons among m alternatives [8].

2.5 Information-theoretic metrics for question selection

Entropy and information gain: In a classification or search context, the Shannon entropy of a set of options T is $H(T) = -\sum_i p_i \log_2 p_i$. Asking question a partitions T and yields conditional entropy $H(T|a)$. The **information gain** from asking a is

$$IG(T,a) = H(T) - H(T \mid a),$$

guiding the choice of the next question to maximize uncertainty reduction [3].

Bayesian expected information gain (EIG): In preference elicitation, the question $d \in D$ has expected information gain

$$\mathrm{EIG}(d) = \mathbb{E}_{p(y|d)}\big[H\big[p(\theta)\big] - H\big[p(\theta \mid y,d)\big]\big],$$

where $p(\theta)$ is the current posterior and $p(y \mid d)$ is the predictive distribution of the user's response. The question with highest EIG is asked next [3].

2.6 Regret-based metrics (optional)

Maximum regret: For each alternative a, define regret under weight vector w as $r(a,w) = u(a^*,w) - u(a,w)$, where a^* is the best alternative given w. The **minimax regret** recommendation is the alternative that minimizes the maximum regret over all feasible weight vectors [9]. Regret-based elicitation asks questions that most reduce the worst-case regret.

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3. Variations and domain-specific considerations

Consideration	Impact on formulas
Qualitative criteria / ordinal data	Require transformation to numeric scales (e.g., mapping "excellent/good/poor" to 1/0.5/0). Outranking methods like ELECTRE handle qualitative data better than WSM/TOPSIS.
Non-compensatory preferences	Weighted sums and TOPSIS assume full compensation. Use ELECTRE or lexicographic rules (veto thresholds) when trade-offs are not allowed [10].
Interactions between criteria	Additive models assume utility independence. If interactions exist, use multiplicative MAUT or multi-attribute value functions.
Uncertain or missing performance data	Bayesian models treat unknown scores as random variables; use imputation or robust ranking (minimax regret).
Large alternative sets	Apply pruning heuristics (veto criteria) and clustering before heavy computation. TTM reduces pairwise comparisons among many alternatives [8].
Dynamic or time-dependent preferences	Preference weights may change over time; incorporate discounting or recency weighting and re-estimate periodically.
Group decisions	Aggregate individual weight vectors via geometric mean (AHP) or social choice functions; check for consensus.

4. Generalized pseudocode for the decision system

Below is high-level pseudocode describing the main stages of the AI decision-support skill. The structure abstracts away implementation details and can be customized to various domains.

```
# Inputs: alternatives A, criteria C, performance matrix X, initial prior
p_theta
# Domain database DB containing attributes of each alternative

def decision_support_system(DB):
    # Stage 1: Values extraction
    objectives = ask_values_questions()
    criteria_subset = map_objectives_to_criteria(objectives)

    # Stage 2: Hard constraint elicitation
    constraints = ask_veto_questions(criteria_subset)
    candidates = filter_db(DB, constraints)
```

```python
    # Stage 3: Adaptive preference elicitation
    # Initialize posterior over preference weights
    posterior = initialize_posterior(criteria_subset)
    while not convergence_condition(posterior):
        # Select next question using EIG or PAPRIKA rule
        question = select_best_question(posterior, candidates)
        answer = ask_user(question)
        posterior = update_posterior(posterior, question, answer)

    # Derive point estimate of weight vector from posterior
    weights = compute_weights_from_posterior(posterior)

    # Stage 4: Computational elimination (optional)
    eliminated = outranking_elimination(candidates, weights)
    remaining = candidates - eliminated

    # Stage 5: Final ranking
    scores = score_alternatives(remaining, weights, method="WSM" or "TOPSIS")
    ranking = sort_by_scores(scores)

    # Stage 6: Explanation and progressive profiling
    explanation = generate_explanation(ranking, weights, constraints)
    store_user_profile(objectives, weights, constraints)
    return ranking, explanation
```

Description of key functions:

- `ask_values_questions()`: Use open-ended prompts to identify fundamental objectives and capture the user's vocabulary (Stage 1).
- `map_objectives_to_criteria()`: Deterministically map extracted values to domain-specific criteria.
- `ask_veto_questions()`: Present lexicographic or fast-and-frugal questions to obtain non-negotiable constraints (Stage 2).
- `filter_db()`: Apply hard constraints to the database to produce a candidate set.
- `initialize_posterior()`: Set an initial prior distribution over preference weights; can be uniform or informed by domain knowledge.
- `select_best_question()`: Choose the next elicitation question using EIG, PAPRIKA or TTM, depending on criterion count and required model (Stage 3).
- `update_posterior()`: Update the posterior distribution or weight estimates based on the user's answer.
- `compute_weights_from_posterior()`: Convert the posterior distribution into a point estimate (e.g., mean or MAP) for scoring.
- `outranking_elimination()`: Run ELECTRE or a similar method to remove dominated alternatives (Stage 4).
- `score_alternatives()`: Compute WSM or TOPSIS scores for the remaining alternatives (Stage 5).

- `generate_explanation()`: Construct a natural-language explanation of the ranking based on weights, constraints and performance.
- `store_user_profile()`: Persist information for progressive profiling and to avoid repetition in future sessions (Stage 6).

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5. Use-case variations

To adapt the formulas and pseudocode to specific domains, consider the following variations:

1. **House selection** – Criteria may include price, square footage, commute time, school rating. Use hard constraints on budget and location; apply WSM/TOPSIS for final ranking; include geographic heuristics (e.g., distance calculations). Missing data may be imputed or flagged for user review.
2. **App design** – Alternatives are sets of feature bundles; criteria may be user engagement, development cost and emotional impact. DOE could be used to simulate outcomes of feature combinations; weights reflect stakeholder priorities. Conjoint analysis may replace WSM to estimate part-worth utilities.
3. **Academic program choice** – Criteria include curriculum match, tuition cost, faculty reputation, location. Hard constraints may include accreditation or funding requirements. TOPSIS is effective when attribute data are quantitative; AHP may be used to elicit weights if criteria are qualitative.
4. **Hiring or vendor selection** – Multi-stakeholder decision; incorporate group AHP or consensus weight aggregation. Outranking with veto thresholds may be necessary to enforce minimum qualifications.
5. **Life decisions (e.g., having children)** – Weight elicitation may not be appropriate; use values clarification, narratives and fast-and-frugal heuristics rather than additive scoring. The system can aid reflection rather than ranking.

Conclusion

This document provides the mathematical foundations and a generalized decision structure for a decision-support skill that can adapt to many domains. Deterministic formulas (WSM, TOPSIS, ELECTRE, AHP) provide transparent and reproducible scoring [1] [2], while information-theoretic and Bayesian expressions guide adaptive questioning to minimize cognitive load [3]. The pseudocode outlines a modular pipeline that integrates these elements, enabling AI agents to customise the framework for specific use cases.

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